

NEWSLETTER

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Meetings are held on the third Monday of the month at 7.30 pm in the Staff Room of Newlands Intermediate, Bracken Road, Newlands, Wellington. Attendance currently averages 10-15 with ideas, information, problems and matters of interest being shared.

Chairman	Jim Oliver	ph 665 205
Secretary/Treasurer	Ross Fletcher	ph 788 454
Librarian	Scotty Baxter	ph 784 619

Public Domain BASIC	Jim Oliver
Public Domain CP/M	Ross Fletcher

NEWSLETTER CONTRIBUTIONS TO: J. Hodgson, 38 Gloaming Hill
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From your Editor

The Editors prize of \$20 this issue goes to Jim Oliver for his endless contributions without which this newsletter would probably be defunct - in this issue helpful explanations of various PD utilities in CP/M but especially info on the Basic INSTR command - thanks Jim.

On the membership front we have culled out those who are overdue prior to Jan 1988. Those more than one month currently overdue will be getting a reminder from the treasurer. The total membership presently current or current at some point in 1988 now stands at some 63. Please check your expiry date on the label on this newsletter as it may be your last if more than one month overdue.

Financially we are reasonably well off with some \$3-400 more than we really require given the lower membership. As we are not a profit making outfit we need to reduce this before it becomes embarrassing - any suggestions would be welcome eg: increase the Editors prize.

The last newsletter was posted on September 27th at Willis Street North P.O. Most people received theirs anything up to 4-6 weeks later. The P.O. has no explanation other than it shouldn't happen. To avoid another such stuff around we have decided to dispense with the hassel of club postage rates (given the now low volume of mags produced) and just use stamps. Maybe they were giving us a hint!

This newsletter is pretty slim as I have run out of material to print. It will be the last issue this year - next issue in the New Year sometime so get your fingers tapping and something in print to me please.

Have a good summer.

.....ed

Explaining the Artist Program

By Michael Warwick

This is a guide to the artist program to get you started (printed in previous issues). I will go into some detail on most of the comands, but first I suggest you (if you haven't already) type in the artist program.

This program was designed to make the best possible use of the keypad on the SVI-328, the only keys which haven't been used are the [select] and [print] keys which can be used by a machine code dump as the one already in the program is rather slow (up to 10 mins for a printout!!).

When you run the art program you will find it takes a little while before it actually starts. This is because it is loading the font into an array, the font is the one you are currently using so special fonts can be used. If you have a font editor just load the font you want to use, and then exit the font editor & run the art program. Any notes relating to the printer are for my Panasonic KX-P1080 which uses the Epson 'standard' and so should run on most other printers.

The program first starts (after loading the font) you will see a small cross in the centre of the screen, in the middle of this cross is a hole and this is where every-thing happens. Press the spacebar. The dot in the middle should turn black. Now move the cross away using the cursor keys and admire that dot. now you can draw things! if you put a dot in the wrong place it can be removed by simply pressing '*'. Now if you want to add colour then press '+' and the cross should change to green (unless you are like me and have an amber monitor or green screen), pressing '-' will return your colour to black. If you press the '+' key enough times the cross will go through all the colours to white then to transparent and black again (if you press '-' the same thing happens in the reverse order). If you find the cross is moving to slow then press ',' and it will go twice as fast. pressing ',' again will make the cross go faster, and faster, until it reaches maximum speed, when it will go back to its old pace.

Now we get into some of the exciting graphics offered by the numeric keys, but first we will clear the screen by pressing '/'. Just one more thing, press the '.' key. This seems to do the same thing as the space, but really it is putting that point into memory for later use. we will start with key '9' and work down to Key '1'.

Key '9': this key paints in an area in the current cross colour. the border of the shape to be filled should be the same colour of the cross or it will just fill the entire screen.

Key '8': draws a box with one corner at the point in memory (where you last pressed '.') and the other at the present cross position. The box is in the present cursor colour.

Key '7': same as '8' except this comand will fill the box (even if there is something in the area where the box is drawn).

Key '6': draws a line from point in memory to the present cross position.

Key '5': draws a circle, with the point in memory as the centre and with the present cross position as the outside.

Key '4': draws a circle as with '5' but then procedes to paint in the circle as with '9'. Unlike the fill done with '7' this paint will not fill in an area inside the circle which has a border the same colour as the cross colour.

Key '3': this function is the very reason I sent this program in for in the first place. Pressing this key disables all of the other functions and puts you into 'type mode'. This means that you can type to the screen in much the same way as I am writing to Wordstar now. Although you don't have all the fancy editing functions you can print in any type-face you can print on the 40 col screen, and a multiple of different sizes. Once you have pressed '3' you can type any thing and it will appear onto the screen. Pressing the up arrow key will make the typing twice as large, and pressing it again will make it even bigger - the only limit is when the size number gets too large and causes a 'overflow' error. When you want the lettering to return to its normal or default size press the down arrow. If you get to the edge of the screen it will automatically go down to the next line.

Key '2': this key photographs a 10x10 area of the screen. The picture that has been taken is printed up in the top right hand corner of the screen.

Key '1': this key prints the photograph taken by '2' onto the screen. If you haven't taken a photo yet then you will get a error message and whatever was on the screen will be lost, never to be seen again! (unless you have saved it of course!). for those of you who are familiar with basic key '2' uses the get command and key '1' uses the put command. This means that it won't always work and quite often funny things can happen, if they do try pressing '1' again, maybe even 2 times. If this doesn't work then you might have to fix it up using the space bar and '*' key.

Pressing 'S' saves your work of art to disk, and pressing 'L' loads a screen from disk. Make sure when you load that you have the right disk in the drive, also be careful when saving that you are not writing over another picture / program of the same name.

Printing your master piece to paper is a slow business so if you can I advise that you use a machine code dump which uses the print key and/or the select key. If you can't do that then there is one built in. Press ^P (CTRL and 'P' at the same time) and then '1' for a small picture or '2' for double size and leave the computer to print it out. The cross shows you where the computer is at and the printing happens each time the cross reaches the edge of the screen.

Remember that colours may begin to 'run' because of the way the screen is stored in memory. You can only have two colours in a 8 pixel long line to put it simply although most of the time this does not matter. If you want more detail try reading 'Peeks and Pokes for the Spectravideo' by Deane Whitmore.

I have a pattern fill which fits nicely on the end of the artist program. It changes the way keys 1,2,4 & 9 work. The pattern fill does not work in the same way as the paint command and if anyone can find a better way to get the same effect please tell me.

To implement the pattern fill add these lines;

```
1000 FOR I = SX TO 0 STEP -1:IF POINT (I,SY)<>A THEN S=I+1 ELSE
--NEXT:S=0
1010 FOR I = SX TO 255:IF POINT (I,SY)<>A THEN E=I-1:RETURN ELSE
--NEXT
1020 E=255:RETURN
1500 A=POINT (SX,SY):GOSUB 1000
1510 FOR I=S TO E:FOR J=SY TO 191
1520 IF POINT (I,J)=A THEN GOSUB 2000:NEXT
1530 FOR J = SY-1 TO 0 STEP -1:IF POINT (I,J)=A THEN GOSUB
--2000:NEXT ELSE J=-1:NEXT J
1540 NEXT I:RETURN
2000 IF C(IMOD8,JMOD8)=1 THEN PSET (I,J),C ELSE PRESET (I,J)
2010 RETURN
```

Then change these lines;

```
20 DIM C(10,10)... BECOMES DIM C(8,8)...
130 ...VAL(A$)> 8 THEN160 BECOMES ...VAL(A$)> 9 THEN170
140 ADD ',1500' TO THE END DELETE 160
580 BECOMES ;580 FOR I=0 TO 7:FOR J=0 TO 7:IF POINT(SX+1,SY+J)=C
--THEN C(I,J)=1 ELSE C(I,J)=0
ADD THIS LINE;585 NEXT J,I:RETURN
590 BECOMES ;590 FOR I=0 TO 7:FOR J=0 TO 7:IF C(I,J)=1 THEN
--PSET(SX+I,SY+J),C
ADD THIS LINE;595 NEXT J,I:RETURN
```

And now you have a pattern fill.

I got the basic idea from an artist program written for the Amstrad (the programmer got 10 Pounds!).

The main problem with this fill is that when you fill most shapes some areas don't get filled, if anyone finds a solution to this problem (besides moving the cross to the problem area and the filling it!) please tell me. I would be most relieved. This fill fills until it reaches a pixel of a different colour to the point where the cross is unlike the paint command which fills till it reaches a pixel of the same colour to the point. The fill is activated by pressing '9' as before. Now '2' only takes a picture of an 8x8 grid and it only recognises dots that are the same colour as the cross. The picture you take with '2' is the pattern used in the pattern fill, and the pattern defaults to a white square.

Key '1' now only puts a 8x8 square to the screen. The pattern now has a white background and the pattern is printed in the colour of the cross.

Now the rest is up to you and your creativity! I'm sure you will get a lot of fun and interest out of it.

now go to it!

....0000000000....

BRAIN TEASER No.2 ANSWER

Jim Oliver

I found this simple problem quite difficult at first. Once I got the hang of it however I realised that there were several different ways of tackling the program. The one I've chosen to give you below is maybe not the fastest or the most elegant but it is the easiest to understand.

Firstly you need to know for speeds sake that our computer carries out arithmetic operations fastest when they are expressed in STRING form. This is convenient because we need to add a "3" to the centre of the top number.

```
10 REM No.11 'Brainteasers' Encyclopedia
20 TIME=0:B$="12456789":T=VAL(B$)
30 B$=RIGHT$(STR$(T),8)
40 TN$=LEFT$(B$,2)+"3"+MID$(B$,3,2):BN$=RIGHT$(B$,4)
50 IF VAL(TN$)=2*VAL(BN$)THEN PRINTTN$,BN$:PRINT(TIME/50)
-- ELSE T=T+1:GOTO 30
```

What we do here is as follows : start off with a number in B\$, note there is no "3". The reason for this is we want the 3 to be in the middle top number. Next we split out the two numbers and add in the "3". We then check to see if the top is twice the bottom and if it is print out the answer and the time taken to get there. If the numbers don't work out then increment T and go back again.

There are ways of checking to see if the number T contains only the numbers we want, i.e no zeros or repeated numbers. However I found that by doing this the program is slowed down further, not speeded up. There are ways of speeding it up ; since (HINT, HINT) the top number must start with "15.."

Have you found a quicker way ? If so let us know.

... Jim Oliver

BRAIN TEASER No. 3

I can only think that the reason we didn't get any replies to Brain Teaser No.2 was that it was too hard. Well here's a much easier one for you. And this time lets have a few replies from the budding programmers out there.

Here's four numbers :
1731
5363
7179
9903

Your job is to write a program to find the LARGEST number that will divide into each of these numbers and give the SAME REMAINDER.

Hint, it won't be any larger than 1731 !

PUBLIC DOMAIN CP/M INFO

Jim Oliver

There are a few CP/M files that I use often. I thought it may be an idea to share these with some of you who may have just started with CP/M and could benefit from some basic info, (I certainly could have)

FBAD56.COM

This COM file is used to check a newly formatted disc for bad sectors or bad system or directory tracks. Any bad sectors found will be "locked out" for general use by assigning them the file name [UNUSED BAD]. As far as CP/M is concerned these areas are now "in use" and will not be used again. This avoids the objectionable BAD SECTOR error (usually in the middle of a large Wordstar or dBase file!)

FBAD56.COM is available in several earlier versions but I always use V.5.6. The syntax for using this file is "FBAD56 D:" where D: is the drive you are testing. I now run this file on all formatted discs including BASIC ones. (However if an error is found on your BASIC disc some funny data is written to track 3 ! in which case you must re format.)

RECLAIM.COM

This is the best of the UNERASE or RECLAIM programs around. Simply type RECLAIM D: where D: is the drive name (if not the default drive) and you are presented with a menu. At the bottom of this is the first of the ERASED files on the disc. You may move to the next ERASED file by pressing <RETURN>. When you get to the file you want to RECLAIM you press "R" then "W" then "Q" (to Quit) What you must realise when using RECLAIM is that that when you ERASE a file all that happens is that the first byte of this files directory entry on the disc is altered from zero to E5 Hex. The actual data is still all there on the disc, PROVIDING you haven't written any new data to it. Even if you have used the disc briefly it's still worth a go for a valuable file. If you find there are heaps of ERASED files on the disc that you are never going to use again it is well worth running the next file; SORTDIR.COM.

SORTDIR.COM

This program checks the directory tracks for ERASED files, then permanently removes these AND rearranges the directory so that it reads alphabetically. I have found this file invaluable for cleaning up Wordstar data discs which after a bit of use can be a real mess. You cannot use RECLAIM.COM after SORTDIR.COM.

DU-VB6.COM and NSWP207.COM

I use both of these files more than any others. NSWP207.COM is version 2.07 of New Sweep and I now have a 12 page DOC file on disc for anyone who is interested. ANYONE who uses CP/M must have at least one version of NSWP for coping, printing, deleting, squeezing (& unsqueezing), renaming, and setting files status.

DU-VB6 is a disc utility which will do virtually ANYTHING you may wish to do to a disc by operating directly on the disc itself. So you have to have a certain minimum amount of knowledge before you use this one. There is a good DOC file available for this one as well.

SVDO.COM

This one is a small (8K) Wordstar NON-DOCUMENT ONLY CP/M editor. This means you can use it for writing letters but it is normally used for writing programs or other files because it does not support word wrap, reformatting or justifying. However it is very fast and does enable you to use nearly all the RAM due to it's small size. There is a 24K DOC file available !! If you want to use MBASIC or SVMBASIC then this editor is MUCH easier to use than the one supported by MBASIC.

WARNING: There are many different versions of this file and nearly all of them have been patched in some way to run on non SV systems. This file is called all sorts of names depending on which computer it is run. However we have the SV version which can be installed to suit your individual requirements.

The "INSTR" Command

Jim Oliver

Have you ever used the command INSTR available in SV BASIC ? You should have since it is a very powerfl and fast way of testing BASIC strings for certain conditions. The syntax used is as follows :

A=INSTR([I,X\$,Y\$)

The variable A returns a number which is the first occurance of the string Y\$ in X\$. The optional I is the starting point for search in X\$ and must be less than 255 since any string <>255 characters.

For example if X\$ = "ABCDEFGHJIMIJKLMNO" and I was trying to find my name in there (Y\$="JIM") then A would return 9. Or if I wasn't sure if my name was in the string or not I could say PRINT MID\$(X\$,INSTR(X\$,"JIM"),3) and if it was there then "JIM" would be printed out.

The INSTR command comes into it's own for use when searching data files for specific information. Data files consisting of records of known length and fields can be searched with speed and ease.

Directing a program to certain lines after inputting from the keyboard is very conveniently done using INSTR. For example

```
10 CM$="AaBbCcXx"
20 PRINT "Please type a letter (A --> C ; X to exit)"
30 A$=INPUT$(1)
40 ON INSTR(CM$,A$) GOTO 50,50,50,50,50,50,70,70
45 BEEP:?"YOU CAN'T FOLLOW INSTRUCTIONS"
50 PRINT"You typed ";A$
60 GOTO 30
70 CLS:PRINT"END"
```

Note that the use of upper or lower case is allowed by the specification in CM\$. The line of 50's can be ANY line number depending on what you want to do.

Recently I wished to test a set of random numbers to for duplication of any numbers as generated by the program. This can be done in several ways but INSTR was the best. If in the above example A\$ is jumbled up in any way then line 120 is "run" on it the answer will always be the same PROVIDED all the numbers are unique. Try it with any number duplicated. Lines 50 - 100 give a random set of numbers BUT with each number used only once.

```
10 INPUT "Number of variables";L:L=L-1:REM Select 9 in this case.
20 IF L=0 THEN BEEP:GOTO 10:REM see notes at end for other no's.
30 DIM N(L):X=RND(-TIME):A$="123456789"
40 INPUT "No. of cycles";C
50 FOR Q=1 TO C
60 FOR J=0 TO L:N(J)=J:NEXT
70 FOR I=0 TO L:J=INT(RND(1)*(L+1))
80 IF N(I)=0 THEN N(I)=9:'for no zero
90 SWAP N(I),N(J)
100 NEXT
110 FOR I=0 TO L:B$=B$+STR$(N(I)):NEXT:PRINTB$;" ";
120 FOR V=1 TO L+1:S=INSTR(B$,MID$(A$,V,1)):SUM=SUM+S:NEXT:PRINT SUM
130 SUM=0:B$=""
140 NEXT Q
```

There is a way of generating the A\$ string automatically without having to define it first as I have done above, can you work it out ?

Notes for Users of Bryan Fisk's Disc Utility

Jim Oliver

I regularly back up the directories of our PD discs whenever add or subtract new files. Sometimes I copy these discs for members using the CP/M COPY.COM file. I have found that some discs which have had the "directory saved as backup" using D Utility do not perform quite as expected.

The copying procedure works OK up to track 20 sector 7, the things go wrong and COPY will not go any further due to BAD SECTOR ! If this disc is checked using the Basic program cal DSKTST, the disc checks out OK, but bombs with CP/M's FBAD.C at group 49H or sector 8. Very strange. (Did you know that FBAD.COM will check a basic disc ?) I can't produce this fail every time but it happens too often to ignore. Maybe this has something to do with DSK0\$ not writing EXACTLY where it should.

Another problem is this : if you select option 6 from the menu (save / load directory) and using a disc which has not yet been backed up, then select the LOAD option in place of SAVE option you will of course load a blank directory (none been saved yet) on top of the current directory tracks. The result of this is the same as a format ! You completely ruin directory. Maybe Bryan can fix this little beauty.